

Atlas

The Agile Humanoid by Boston Dynamics

| Overview

Atlas is widely considered the world's most dynamic humanoid robot, designed and developed by Boston Dynamics. Initially funded by DARPA, Atlas has evolved significantly over the years, transitioning from a tethered, heavily hydraulic machine to a fully electric, incredibly agile robot capable of complex parkour, backflips, and sophisticated object manipulation.

| Key Capabilities

Atlas pushes the limits of whole-body mobility and bimanual manipulation. Its advanced control system enables highly dynamic movements, maintaining balance even when shoved or while navigating uneven terrain.

- **Agility:** Capable of gymnastics, parkour, and running.
- **Control System:** Uses model predictive control and complex sensor fusion to plan and execute motions in real-time.
- **Recent Evolution:** The latest generation has moved away from hydraulics to custom, high-power electric actuators, offering a broader range of motion and increased strength.

Technical Focus

Unlike commercial-first robots, Atlas has largely served as a research and development platform to push the boundaries of what robotic hardware and control algorithms can achieve. Its hardware is custom-built to maximize power-to-weight ratios.